(10)

~~Ex:-24~~:- (or)

By using definition of distribution.

New person weight = ~~Replaced~~ Actual weight + No of observations × ↑/↓ in Avg

$= 65 + 8 \times 2.5 = 65 + 20 = \boxed{85 \text{ kgs}}$

Ex:-25:- [B.Q:-15]:- A man said to his son, "I was $\frac{2}{3}$rd of your present age when you were born". If the present age of the man is 48 yrs. Find the present age of son?

Ans:- Let's assume present age of son = S

when son was born (son age yr's ago) means 'S' yr's ago.

$48 - S = \frac{2}{3}S \Rightarrow 144 - 3S = 2S$

$\Rightarrow 5S = 144 \Rightarrow \boxed{S = 28.8 \text{ yrs}}$

Ex:-26:-[B.Q:-16]:- Present ages of x & y in the ratio 5:6. 7 yrs hence this ratio will become 6:7 respectively. what is x's present age?

Ans:- $x:y = 5:6 \Rightarrow x = 5a, y = 6a.$

7 yrs hence $\Rightarrow (x+7):(y+7) = 6:7$

$\Rightarrow \frac{5a+7}{6a+7} = \frac{6}{7} \Rightarrow 35a + 49 = 36a + 42 \Rightarrow \boxed{a = 7}$

x's present age $= 5a = 5 \times 7 = 35$ yr's

(or)

When $+1\left(\begin{matrix}5:6\\6:7\end{matrix}\right)+1$ means hence yr's (7) only each part value.

x age $= 5a = 5 \times 7 = 35$, y age $= 6a = 6 \times 7 = 42$

Ex:-27:-[B.Q:-17]:- 6 yrs ago, the ratio of ages of Kunal & Sagar was 6:5. 4 yr's hence, the ratio of their ages will be 11:10. what is Sagar's Present age?

Ans:- 6 yr's ago ratio $= k-6 : s-6 = 6:5 \Rightarrow k-6 = 6a, s-6 = 5a$

$\Rightarrow k = 6a+6, s = 5a+6 \quad \text{---} (1)$

(11)

4 yr's hence their ratio = $k+4 : 5+4 = 11:10$ — (2)

Sub (1) in (2) $\Rightarrow \frac{6a+6+4}{5a+6+4} = \frac{11}{10}$

$\Rightarrow 60a+100 = 55a+110 \Rightarrow 5a = 10 \Rightarrow a = \underline{2}$

Sagars present age $= S = 5a+6 = 5\times 2+6 = \underline{16 \text{ yr's}}$

Ex:-28:- [B.Q:- 18]:- I went to a hotel along with 12 friends. I paid Rs. 145/- all the others paid an equal amount. In the end we did some calculations & found that avg paid by all of us was Rs. 5/- more that what was originally paid by each of my friends. How much money did each of my friends pay?

Ans:- 12 frnds paid an equal assume $= x \Rightarrow (\text{sum}) = 12x$

Avg of 13 $= x+5$.

Sum of 13 people $= 12x + (12 \text{ sum}) + 145$

Avg of 13 $= x+5 = \frac{12x+145}{13}$

$\Rightarrow 13x+65 = 12x+145 \Rightarrow \boxed{x = 80}$

(or) By using definition:-

C.A $= (x+5) =$ Extra person − No of person other than from x ped / ed value after addition.

$\Rightarrow x+5 = 145 - 12\times 5 = 85 \Rightarrow \boxed{x = 80}$